From precision to strength: computer vision for suture quality assessment-an ex vivo pilot study.

Authors: Spagnulo Roberto, Marzola Francesco, Corso Federica, Distefano Giovanni, Pescio Matteo, Hayashi Kengo, Barontini Federica, Dagnino Giulio, Althoefer Kaspar, Siciliano Bruno, Ourselin Sebastien, Arezzo Alberto

Journal: Surgical endoscopy | **Year:** 2026 | **DOI:** 10.1007/s00464-025-12441-6

Link: https://doi.org/10.1007/s00464-025-12441-6 (open with institutional access)

Abstract

Suturing is the cornerstone of surgical practice, yet its assessment continues to rely on subjective evaluation. As minimally invasive techniques become increasingly central in surgery, the demand for precision grows. In this context, the present pilot study aims to investigate whether spatial metrics of suture placement can quantitatively reflect mechanical performance and to determine how operator experience and surgical platform influence technical outcomes. Fifteen participants, stratified by prior experience in minimally invasive surgery, performed standardized suturing tasks on porcine rectal specimens using three platforms: conventional laparoscopy, the daVinci Research Kit (dVRK), and the Flex robotic endoscope. Quantitative Suture Assessment was conducted by extracting spatial features (e.g., distance variability between consecutive stitches), while mechanical resistance was evaluated via intraluminal burst pressure. Statistical analyses included correlation and regression modeling to assess the relationship between spatial metrics and burst pressure, as well as comparative analyses across platforms and experience levels. Mean burst pressures measured 17.38 ± 6.54 mmHg for laparoscopy, 15.99 ± 7.69 mmHg for dVRK, and 13.60 ± 10.08 mmHg for Flex. Masters recorded a mean burst pressure of 17.95 ± 9.43 mmHg, while Advanced 13.89 ± 7.41 mmHg and Beginners 15.13 ± 7.64 mmHg. Laparoscopy achieved higher burst pressures than Flex, and dVRK outperformed Flex among Beginners. Laparoscopy and dVRK were faster than Flex, with Masters completing tasks more rapidly across all platforms. Spacing irregularity was negatively correlated with burst pressure ($p < 0.05$). However, the XGBoost model trained on all variables exhibited poor performance ($R2 = -0.42$, MSE = 93.7), and high multicollinearity. Platform-specific scores emerged with limitations. While spatial metrics correlate with mechanical resistance, they appear insufficient as standalone indicators of suture quality. Ultimately, this approach could pave the way toward intelligent, intraoperative systems capable of delivering real-time feedback, with implications for both surgical education and quality assurance.

Note: Publisher PDF could not be downloaded from this network (publisher access restrictions). This record provides the full citation and abstract; obtain the full text via the DOI link above.